Quantifying the effects of nickel on Earth's inner-core nucleation

Jiahui Zhai[1#], Liangrui Wei[1#], Chen Gao[1], Yang Sun[1,*]

[1]Department of Physics, Xiamen University, Xiamen 361005, China

Key Points

- Nickel-enriched liquid can significantly reduce the undercooling required for inner-core formation.
- The body-centered cubic structure nucleates more easily than stable hexagonal close-packed Fe-Ni under core conditions.
- Chemical variations in the liquid core are key to reducing the supercooling needed to start inner-core growth.

Abstract

The formation of Earth's solid inner core marks a major transition in the thermal and chemical evolution of the deep Earth, yet its origin remains paradoxical, as initial nucleation appears to require unrealistically large supercooling in the outer core. Here, we use atomistic simulations to quantify how Ni affects this process under inner-core conditions. While Fe-Ni alloys preserve strong thermodynamic competition between the hcp and bcc phases, the bcc phase consistently forms smaller critical nuclei and has lower nucleation barriers than hcp. Increasing Ni content in the melts further lowers the nucleation barrier and shortens the nucleation waiting time. Local chemical fluctuations also strongly affect the macroscopic nucleation rate. Combining these effects, bcc nucleation in $Fe_{80}Ni_{20}$ reaches about 250 K of supercooling, approaching geophysical constraints. We demonstrate that Ni enrichment, bcc nucleation, and chemical heterogeneity substantially narrow the inner-core nucleation paradox.

[#]These authors contributed equally.
[*]Email: yangsun@xmu.edu.cn

Plain Language Summary

Earth’s inner core began to grow when liquid metal at the planet’s center first formed stable solid nuclei. Earth’s inner core began to grow when liquid metal at the planet’s center first formed stable solid nuclei. This event is significant for Earth’s evolution because inner-core growth releases heat and chemical energy that help sustain outer-core convection, which is key to maintaining the global geomagnetic field. However, theoretical models suggest that iron, the dominant element in the core, requires much more cooling to nucleate than the liquid core could realistically provide, creating the inner-core nucleation paradox. We studied how nickel, the second most abundant metallic element in the core, affects this process. Using atomistic simulations, we found that iron-nickel liquids can form an initial body-centered cubic solid more easily than the usual hexagonal close-packed form. Higher nickel content lowers the barrier to forming the first solid, and spontaneous nickel-rich fluctuations in the liquid make nucleation more accessible. Under favorable iron-nickel compositions, these effects reduce the required cooling to values much closer to geophysical estimates. Nickel and chemical heterogeneity therefore appear to be important factors in explaining how Earth’s inner core started to crystallize.

Keywords

Earth's core; nucleation; Fe-Ni alloy

1. Introduction

The solidification of the Earth's inner core (IC) provides a key energy source to sustain the geodynamo through the release of latent heat and the partition of light elements (Lister & Buffett, 1995; Sumita & Bergman, 2015). Although seismology and mineral physics constrain many present-day properties of the inner core (Birch, 1988; Deuss, 2014; Fischer & McDonough, 2025; Hirose et al., 2013, 2021), the mechanism that initiated its solidification remains an open question. Thermal evolution models usually assume that nucleation begins once the core temperature falls below the melting point (Nimmo, 2015). However, under the geologically long-time equilibrium of the liquid core, homogeneous nucleation requires additional undercooling to overcome the barrier caused by the formation of the solid-liquid interface (SLI). Calculations with classical nucleation theory (CNT) suggest that the nucleation of the stable hexagonal close-packed (hcp) iron phase requires substantial undercooling far larger than the plausible value that core can reach with the secular cooling, leading to the inner core nucleation paradox (Huguet et al., 2018; Wilson, Davies, Walker, Pozzo, et al., 2025).

The paradox has been explored by investigating intrinsic nucleation pathways in pure Fe. Molecular dynamics (MD) simulations show that crystallization can proceed through a kinetically favored two-step pathway mediated by a metastable body-centered cubic (bcc) phase (Y. Sun et al., 2022). This scenario is consistent with Ostwald's step rule, which predicts that a metastable polymorph can serve as a kinetic gateway, as observed in many materials (Karthika et al., 2016; Rein Ten Wolde & Frenkel, 1999). In metallic systems, bcc often has a lower solid-liquid interfacial energy than close-packed structures (Herlach, 2001; D. Y. Sun et al., 2004). Although hcp remains the stable phase of Fe at IC pressures (Anzellini et al., 2013; Kraus et al., 2022; Tateno et al., 2010), *ab initio* and machine-learning simulations indicate that bcc becomes energetically competitive near melting (Belonoshko et al., 2017, 2021; González-Cataldo & Militzer, 2023; Li & Scandolo, 2024; Y. Sun et al., 2023; Vočadlo et al., 2003; Yang et al., 2026; Zhang et al., 2025). Recent experiments also suggest the transient bcc formation during thermal cycling close to inner-core conditions (Konopkova et al., 2025). These results suggest that bcc-mediated two-step nucleation is likely relevant to the onset of inner-core solidification (Gao et al., 2025).

In addition to pure Fe, the core also contains Ni and several light elements (McDonough, 2003). Previous studies have considered how Fe alloying with light elements of S, Si, O, and C modifies its nucleation (Davies et al., 2019; Wilson, Davies, Walker, & Alfè, 2025; Wilson et al., 2023;

Wilson, Pozzo, et al., 2025). These studies showed that while elements such as C and O can lower the nucleation barrier, their addition simultaneously depresses the melting point. In contrast, Si and S severely hinder the nucleation process, significantly increasing the required undercooling. By contrast, the role of Ni has not been quantified, despite its abundance in the Earth's core, with estimated concentrations ranging from 5% to 15% based on cosmochemical and geochemical models (Birch, 1988; Buchwald, 1975).

Ni has been shown to play a critical role in influencing the thermodynamic properties of Fe under core conditions. Experiments of FeNi alloy suggests possible formation of bcc phase near the core conditions(Dubrovinsky et al., 2007; Ikuta et al., 2021). *Ab initio* simulations suggested that Ni can increase the bcc stability in the FeNi alloys (Chatterjee et al., 2021; Y. Sun et al., 2024; Wei et al., 2025). It is also shown that Ni has a much higher melting point than Fe under IC conditions (Pereira et al., 2025), unlike the light elements that mostly only reduce the Fe's melting point (Wilson et al., 2023). Together, these data imply that Ni may influence both phase competition and nucleation kinetics in a way that is distinct from light-element alloying.

In this work, we quantify how Ni affects the nucleation of Fe-Ni alloys under Earth's inner-core conditions. We combine the thermodynamic calculations with nucleation simulations to determine the critical nucleus sizes and the free-energy barriers and for both bcc and hcp nucleation in the FeNi alloy. We further analyze spontaneous compositional fluctuations in the FeNi liquid and evaluate how they modify the effective nucleation rate. We aim to connect phase stability, nucleation pathway selection, and local chemical fluctuations during the initial crystallization of the Earth's inner core.

2. Methods

We calculated Gibbs free energies of liquid, hcp, and bcc Fe and Ni using Gibbs-Helmholtz integration from the melting point obtained by solid-liquid coexistence simulations (Morris et al., 1994; Y. Sun et al., 2022). Alloy free energies were computed with a regular solution model (Wei et al., 2025).

Critical nucleus sizes were determined with the persistent-embryo method (PEM), which facilitates nucleation simulations with an adaptive potential without biasing the critical region (Y. Sun et al., 2018). Nucleation rates were obtained from steady-state nucleation theory using atomic attachment rates computed by the iso-configurational ensembles (Widmer-Cooper et al., 2004). Additional simulation details and equations are provided in the Supplementary Information.

All molecular dynamics simulations were performed in LAMMPS (Thompson et al., 2022) with an embedded-atom Fe-Ni potential refined from our previous work to better reproduce first-principles interactions (Wei et al., 2025). Simulations used the NPT ensemble with the Nosé-Hoover thermostat and barostat (Hoover, 1985). The MD timestep was 1 *fs*. For the PEM simulations, the system size was set to ~ 53000 atoms. Local structures were identified with cluster alignment (Y. Sun et al., 2016) and polyhedral template matching (Larsen et al., 2016) and thresholds from the equal-mislabeling criterion (Espinosa et al., 2016). During nucleation simulations, a small fraction of interfacial atoms occasionally exhibited fcc-like order within the hcp phase, which are classified as stacking-fault structures of the hcp phase.

3. Results

3.1 Fe-Ni Phase Diagram and Free Energy

We first compare the free energy and phase diagram of the Fe-Ni system calculated using the present EAM potential with previous DFT results (Wei et al., 2025). Figure 1(a) shows the comparison of relative Gibbs free energy potential referenced to the liquid phase between the current EAM potential and DFT calculations. Near the melting points, the deviations in ΔG^{S-L} are only a few meV/atom, and the corresponding shifts in the endmember melting temperatures are less than 70 K. Thus, the present potential accurately describes the thermodynamics of both liquid and solid Fe and Ni endmembers. In Fig. 1(b), the EAM potential predicts a liquid mixing enthalpy that agrees well with the DFT results. The mixing enthalpy of the hcp phase is also well reproduced, with differences of less than 5 meV/atom compared to the DFT data. It also correctly captures the relative ordering of the mixing enthalpies of the bcc, hcp, and liquid phases. However, the potential does not reproduce the negative mixing enthalpy of the bcc phase at ~5 at.% Ni and its rapid increase at 10–15 at.% Ni. This trend of mixing enthalpy is a deviation from regular-solution behavior and indicates anomalous ordering in the Fe–Ni alloy, analogous to the inversion of short-range order reported in Fe–Cr alloys (Caro, 2005; Mirebeau et al., 1984). Reproducing such a low-concentration anomaly in Fe–Cr required an explicitly concentration-dependent EAM potential (Caro, 2005), which is substantially more complicated than the present form. These differences in free energy cause the Fe-Ni phase diagram in Fig. 1(c) to exhibit a larger bcc stability region and higher liquidus temperature by ~100 K in the 5-8 at.% Ni range compared to the *ab initio* phase diagram in (Wei et al., 2025). Nevertheless, the key features of the phase diagram are preserved. In particular, the liquidus temperature increases with increasing Ni composition, indicating that Ni stabilizes the solid phases relative to the liquid under IC conditions. The phase diagram also retains the coexistence of the bcc and hcp phases near the liquidus, capturing the competition between these phases with increasing Ni content. The potential further shows good agreement with DFT results for the liquid structures of Fe, Ni, and Fe–Ni alloys, as shown in Supplementary Material Fig. S1. We therefore employ the present potential to investigate the nucleation behavior of Fe–Ni alloys.

The free energy data computed via the EAM potential allows us to compute the nucleation driving force, $\Delta\mu$. For pure Fe, the nucleation driving force is the same to the Gibbs free energy between solid and liquid, i.e., $\Delta\mu_{Fe} = \Delta G_{Fe}^{s-l}$ (Davies et al., 2019; Y. Sun et al., 2022). In a liquid solution, the nucleus can have a different composition with the liquid. Thus, the nucleation driving force should be defined as the chemical potential change associated with the phase transition as illustrated in Fig. 1(d), expressed as

$$\Delta\mu(x_s, x_l) = G^s(x_s) - \left[G^l(x_l) + (x_s - x_l) \left. \frac{\partial G^l(x)}{\partial x} \right|_{x_l} \right] \quad (1)$$

where $G^l(x_l)$ and $G^s(x_s)$ are the Gibbs free energies of the liquid and solid at compositions of x_l and x_s, respectively (Volkmann et al., 1997).

3.2 Nucleation in Fe-Ni Solutions

To probe the nucleation kinetics, we perform PEM simulations for $Fe_{90}Ni_{10}$ liquid at 5157, 5284, and 5384 K. The initial Ni content of the embryo is set between 5 and 15 at.%. Representative PEM trajectories are provided in Supplementary Material Figs. S2 and S3 for the bcc and hcp phases, respectively. Unlike the single critical nucleus observed in pure Fe and Ni systems (Y. Sun et al., 2018, 2022), the critical nuclei formed in the binary liquid solution can exhibit significantly different sizes and compositions while still reaching the critical region. Complete statistics on the size and composition of the critical nuclei for the hcp and bcc phases obtained from the PEM simulations are shown in Fig. 2(a) and (b). The critical nuclei exhibit a wide range of Ni compositions and sizes, with compositions centered around that of the liquid. At a fixed liquid composition, nuclei with higher Ni content tend to have smaller critical sizes. The nucleus size also decreases systematically with decreasing temperature. Comparison between the bcc and hcp results further shows that the bcc phase consistently has a smaller critical nucleus size than the hcp phase at the same temperature.

While the nucleus size depends on its composition, the nucleation barrier, ΔG^*, obtained from $\Delta G^* = \frac{1}{2}|\Delta\mu|\, N^*$, is nearly independent of nucleus composition, as shown in Figs. 2(c) and 2(d). Thus, compositional differences in the nucleus primarily affect its size but do not significantly alter the nucleation barrier. Comparison between Figs. 2(c) and 2(d) further indicates that the bcc phase consistently has a lower nucleation barrier than the hcp phase across all temperatures. A similar trend, with composition-dependent nucleus size but composition-independent nucleation barrier, is also observed in the $Fe_{80}Ni_{20}$ liquid, as shown in Supplementary Material Fig. S4. These results indicate that the nucleation barrier in Fe-Ni alloys is primarily controlled by the liquid composition. For instance, the $Fe_{80}Ni_{20}$ liquid systematically exhibits a nucleation barrier approximately 20% lower than that of $Fe_{90}Ni_{10}$ for both hcp and bcc phases.

Figure 3(a) and (b) show the nucleation barrier ΔG^* and rate J computed by $J = \kappa \exp\left(-\frac{\Delta G^*}{k_B T}\right)$, where κ is a kinetic prefactor. The bcc phase exhibits a consistently lower barrier than hcp. The addition of Ni significantly reduces the barrier for both bcc and hcp. For instance, at $T = 5400$ K, the hcp nucleation barrier decreases by about 20 $k_B T$ when comparing pure Fe to $Fe_{90}Ni_{10}$. These differences of nucleation barrier translate into orders-of-magnitude variations in the nucleation rate J, as illustrated in Fig. 3(b). Higher Ni composition and formation of the bcc phase dramatically increase the nucleation rate. For instance, at $T = 5400$ K, the fastest rate of bcc nucleation in the $Fe_{80}Ni_{20}$ liquid is about 10^{35} times higher than hcp nucleation rate in pure Fe liquid. For all nucleation-rate data with the same composition x in Fig. 3(b), the temperature dependence can be well fitted by the empirical formula (Bokeloh et al., 2011),

$$J(x,T) = \Gamma(x) \exp\left\{-\frac{B(x)T^2}{[T - T_m(x)]^2}\right\}, \quad (2)$$

where $T_m(x)$ is the composition-dependent melting temperature, and $\Gamma(x)$ and $B(x)$ are fitting parameters. We further describe the compositional dependence of the fitted $\Gamma(x)$ and $B(x)$ using quadratic functions in Supplementary Material Table S1, which provides an interpolation of $J(x,T)$. The resulting $J(x,T)$ curves, plotted in Fig. 3(c), agree well with the calculated raw data.

Given that the nucleation rate is highly sensitive to Ni concentration, it is essential to account for compositional fluctuations in the liquid arising from thermal fluctuations or structural inhomogeneities. To quantify these fluctuations in a stable liquid without convection, we performed large-scale MD simulations of Fe-Ni alloys containing 2×10^6 atoms for $5\,ns$. The simulation cell was then divided into $10 \times 10 \times 10$ subdomains, each containing approximately 2000 atoms. The local Ni fraction within each sub-volume is computed to construct the probability density $P(x_{loc} \mid x_l)$, following (Wilson, Pozzo, et al., 2025). The resulting local Ni concentration distributions are well described by Gaussian statistics (Fig. 3c, inset). By accounting for compositional non-uniformity, the macroscopic nucleation rate can be calculated as a spatial average of local contributions from regions with different compositions, expressed as $\bar{J} = \frac{1}{V}\int J(c)dV(c)$ (Yi et al., 2020). By invoking spatial ergodicity, the volume fraction $\frac{1}{V}dV(x_{loc})$ occupied by transient regions with a specific local composition is statistically equivalent to the probability density $P(x_{loc}|x_l)dx_{loc}$ of encountering such a region. Accordingly, we formulate the macroscopic nucleation rate including compositional fluctuations, J_m, as

$$J_m = \int_0^1 J(x_{loc})P(x_{loc} \mid x_l)\, dx_{loc} \tag{3}$$

where $P(x_{loc}|x_l)$ is the probability density of finding a local composition x_{loc} within a bulk liquid characterized by a mean composition x_l. $J(x_{loc})$ is the nucleation rate described by Eq. (2), assuming a uniform compositional distribution within the local volume. As shown in Fig. 3(d), incorporating these fluctuations via Eq. (3) systematically increases the nucleation rate by orders of magnitude, particularly at higher temperatures.

3.3 Inner-Core Nucleation

IC nucleation is triggered only after the liquid becomes supercooled below the melting temperature (Huguet et al., 2018). The allowable range of supercooling can be constrained from both mineral-physics calculations and geophysical observations (Wilson, Davies, Walker, Pozzo, et al., 2025). Mineral-physics constraints are inferred from the difference between the core temperature profiles and the melting curves, giving available supercooling estimates that range from 200 K (Huguet et al., 2018) to an upper bound of 420 K (Wilson et al., 2023). Geophysical constraints provide a more restrictive estimate assuming that a supercooled core would undergo rapid initial freezing, leaving trapped liquid within the IC (Lasbleis et al., 2020). Linking the volume of trapped liquid (Pang et al., 2023; Singh et al., 2000) to the initial rapid growth suggests a supercooling of less than 100 K (Wilson, Davies, Walker, Pozzo, et al., 2025). These constraints are not fully independent, because the inferred nucleation condition, latent heat release, and subsequent IC growth are coupled. Fully resolving the apparent discrepancy between the ~420 K mineral-physics upper bound and the <100 K geophysical estimate requires a comprehensive model coupling core composition, latent heat, and trapped liquid, which is beyond our present scope. We adopt the broader range up to 420 K as a compatible supercooling estimate (Wilson, Davies, Walker, Pozzo, et al., 2025).

To compare with these supercooling constraints, Figure 4(a) converts the calculated nucleation rates of FeNi alloy into nucleation waiting times, defined as $\tau_v = 1/2J$ (Davies et al., 2019; Y. Sun et al., 2022). Assuming a nucleation incubation time on the order of 1 Gyr and a nucleation volume of ~7.6×10^{18} m^3 i.e., the present-day IC volume, we estimate possible core nucleation waiting time, $\tau_{\oplus}$, to be on the order of 10^{35} $m^3\cdot s$, as indicated in Fig. 4. For pure Fe, hcp nucleation requires unrealistically deep supercooling to meet this timescale. In contrast, the metastable bcc nucleation substantially reduces the waiting time, shifting the required supercooling to smaller magnitudes. This highlights the central kinetic role of bcc formation (Y. Sun et al., 2022).

Alloying with Ni further accelerates nucleation. At a given supercooling, both $Fe_{90}Ni_{10}$ and $Fe_{80}Ni_{20}$ exhibit shorter waiting times than pure Fe, and this effect becomes stronger with increasing Ni content. In particular, bcc nucleation in $Fe_{80}Ni_{20}$ liquid shifts the required supercooling close to the <420 K mineral-physics constraint, reflecting the combined effects of bcc formation and reduced nucleation barriers in Ni-rich liquids. Furthermore, incorporating compositional fluctuations in Fig. 4b reduces the required supercooling by another ~100 K, bringing it into the ~250 K regime. Together, these data indicate that bcc formation, Ni-rich melts, and local compositional fluctuations act cooperatively to facilitate IC nucleation.

4. Discussion

Our findings support initial bcc nucleation as a kinetic gateway that bypasses the high nucleation barrier for hcp formation. This mechanism was first established for pure Fe, where the metastable bcc phase was found to have a lower nucleation barrier in the melt (Y. Sun et al., 2022). Comparative molecular dynamics studies using different interatomic potentials have further confirmed this mechanism under IC conditions (Gao et al., 2025). However, quantitative estimates of the waiting time differ among simulations using different interatomic potentials (Gao et al., 2025). For instance, PotM, developed by (Y. Sun et al., 2022), gives a required supercooling of 610 K for hcp nucleation in pure Fe, whereas simulations using PotA, developed by (Davies et al., 2019), give a required supercooling of 820 K (Gao et al., 2025). This difference can be traced to the *ab initio* data used to develop these interatomic potentials. PotM was fitted using the PAW8 potential, which gives an Fe melting temperature of 5858 K at 323 GPa. In contrast, PotA shows an Fe melting temperature of 6215 K at 323 GPa, closer to the *ab initio* value of 6357 K obtained using the more accurate PAW16 potential for describing the electronic structure at the same pressure (Y. Sun et al., 2023). The present work addresses this discrepancy by fine-tuning the free energy of the EAM potential (Wei & Sun, 2026), yielding an Fe melting temperature of 6354 K in close agreement with the PAW16 result. As shown in Fig. 4, the present waiting time for hcp nucleation in pure Fe is therefore consistent with that obtained using PotA, allowing the effect of Ni on Fe nucleation to be compared more directly with PotA-based estimates (Wilson, Davies, Walker, & Alfè, 2025; Wilson et al., 2023).

The present results emphasize the critical role of Ni in IC nucleation. Cosmochemical models and studies of iron meteorites suggest that Earth's core contains approximately 5–15% Ni (Dubrovinsky et al., 2007; Hirose et al., 2021; McDonough, 2003). Within this compositional range, Ni substantially reduces the waiting time and shifts the required supercooling toward the 600–400 K regime, as shown in Fig. 4(b). This range approaches the upper bound allowed by mineral-physics constraints (Wilson et al., 2023), although it remains above the more restrictive geophysical estimate of <100 K (Wilson, Davies, Walker, Pozzo, et al., 2025). Previous work on Fe-light-element alloys suggested that C can also accelerate nucleation, but more than 11 at.% C, corresponding to 2.6 wt.%, would be required to achieve nucleation at ~400 K supercooling (Wilson, Davies, Walker, & Alfè, 2025; Wilson et al., 2023). Such a carbon content is much higher than recent geochemical estimates of ~0.2 wt.% C in the liquid core from metal-silicate partitioning experiments (Fischer et al., 2020). A coupled effect between Ni and C may therefore provide a possible route to simultaneously satisfying nucleation and geochemical constraints.

Another important factor revealed here is that compositional variations in Fe-Ni liquid can strongly affect nucleation. The local compositional fluctuations considered here only arise from equilibrium thermal fluctuations in a chemically homogeneous liquid and therefore represent the smallest-scale contribution to this effect. In the liquid core, however, compositional heterogeneity is unlikely to be limited to local equilibrium fluctuations. The geomagnetic field has persisted for much of Earth's history, with paleomagnetic records suggesting an active geodynamo as early as the Hadean, implying long-lived convection in the liquid core well before IC nucleation (Tarduno et al., 2020). Such convection could have continually advected and reorganized chemical heterogeneity in the liquid core. Numerical models of compositional convection show that chemical plumes and blobs can persist over long timescales because compositional anomalies diffuse much more slowly than thermal anomalies (Bouffard et al., 2019). Recent seismic observations further suggest that the outer core may be far from well mixed on decadal timescales, with localized transient flows that may be chemically enriched (Zhou, 2022). These processes could broaden the distribution of local compositions sampled by nucleation events beyond the equilibrium fluctuations captured in our atomistic simulations. Chemical stratification provides another geophysically important source of compositional heterogeneity. Recent studies have suggested that the outer core may contain chemically stratified regions, including light-element stratification near the core-mantle boundary and compositional inhomogeneity in the lowermost outer core (Gan et al., 2026; Ganguly, 2025; Gubbins & Davies, 2013). Although these studies mainly concern light elements rather than Ni, they demonstrate that the liquid core may not be compositionally uniform at length scales relevant to core evolution. In such a chemically heterogeneous liquid, stratification and convection could modify the background composition sampled by nucleation events, while thermal fluctuations broaden the local composition around this background. Because the nucleation rate is highly sensitive to composition, these processes may increase the probability of transient Ni-rich or otherwise chemically favorable regions, further reducing the required supercooling and promoting initial IC nucleation. Therefore, during IC formation, the liquid core may have sampled a broad and dynamically evolving compositional field. A natural next step toward addressing the paradox of IC nucleation is to couple Fe-Ni-light-element crystallization kinetics with long-term models of chemical heterogeneity.

In summary, we quantified how Ni affects Fe nucleation under Earth's inner-core conditions by combining free-energy calculations with persistent-embryo simulations. We find that inner-core nucleation is favored by a metastable bcc pathway and that increasing Ni content further lowers the nucleation barrier and reduces the required supercooling. When compositional fluctuations are included, bcc nucleation in $Fe_{80}Ni_{20}$ reaches the ~250 K supercooling regime, substantially narrowing the gap between atomistic nucleation kinetics and geophysical constraints on inner-core formation. These results suggest that Ni enrichment and chemical heterogeneity are important ingredients in resolving the inner-core nucleation paradox.

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Data Availability Statement: The authors comply with the AGU's data policy, and the datasets of this paper are available at Zenodo repository.

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FIG. 1. Gibbs free energy and phase diagram of the Fe-Ni alloy at 323 GPa. (a) Relative Gibbs free energy referenced to the liquid phase. The DFT data are taken from (Wei et al., 2025). (b) Mixing enthalpies, ΔH_{mix}, of hcp, bcc, and liquid Fe-Ni solutions at 6000 K. The lines denote fits to the regular solution model. (c) Fe-Ni phase diagram (d) Calculation of the nucleation driving force for a nucleus with composition x_S emerging from a liquid with composition x_l. The dashed line is tangent to the liquid free-energy curve.

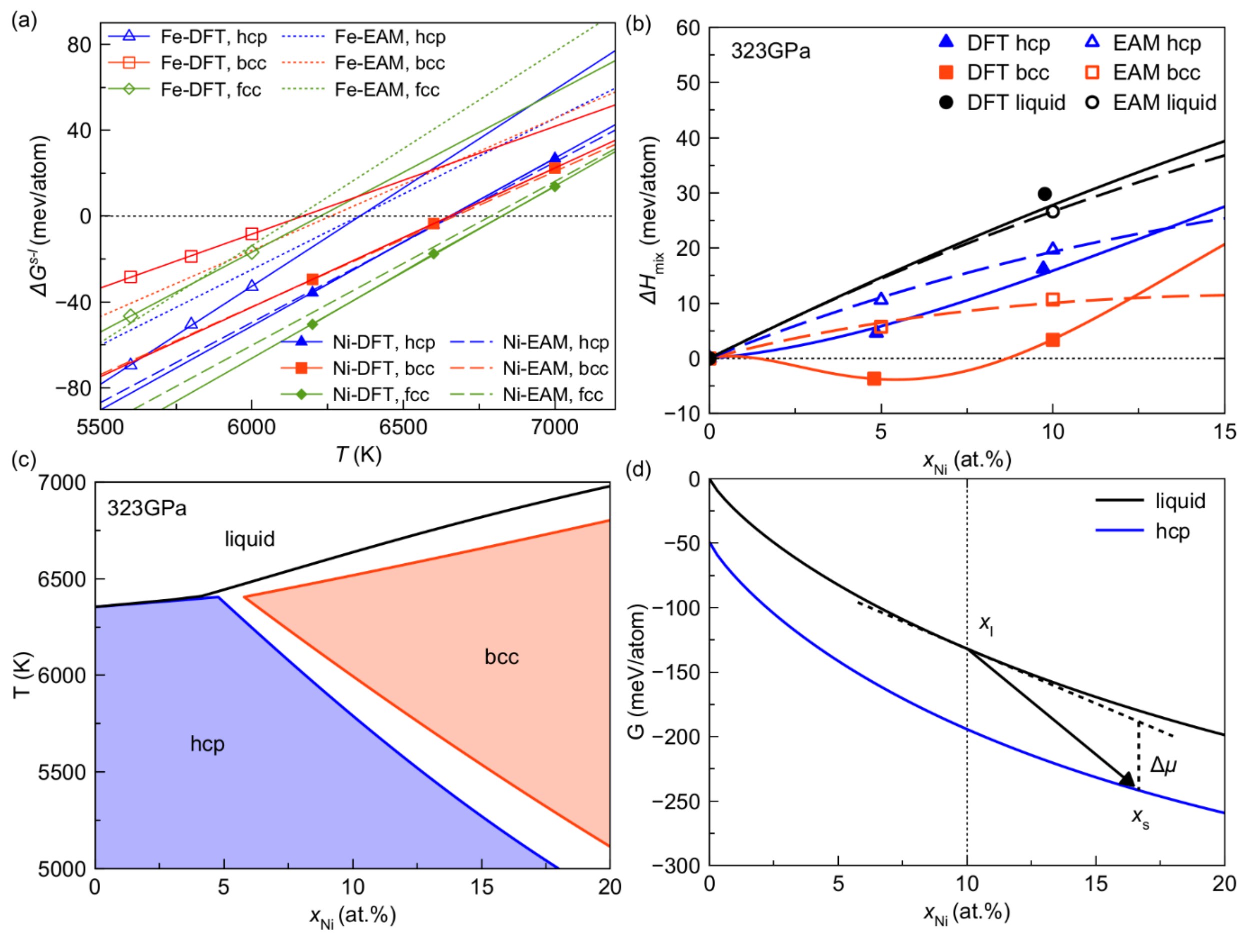


FIG. 2. Critical nucleus size and nucleation barrier of $Fe_{90}Ni_{10}$ liquid at 323 GPa. Size and composition of the critical nuclei for the (a) hcp and (b) bcc phases. The dotted lines indicate linear fits. Nucleation free-energy barriers for the (c) hcp and (d) bcc phases.

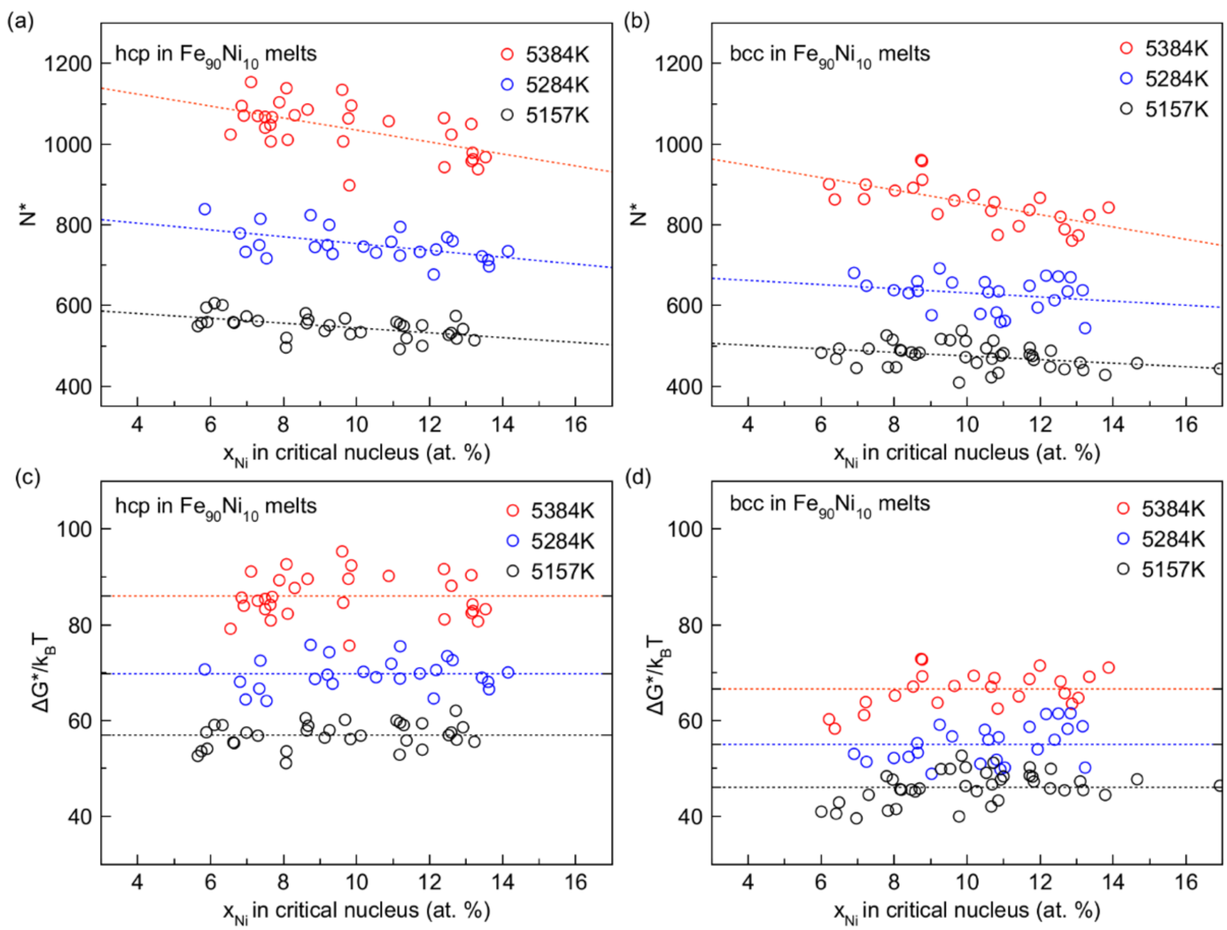


FIG. 3. Nucleation barrier and nucleation rate of Fe-Ni alloys. (a) Nucleation barrier versus temperature for pure Fe (green), $Fe_{90}Ni_{10}$ (blue), and $Fe_{80}Ni_{20}$ (red). (b) Nucleation rate versus temperature. Dashed and solid lines are fits to Eq. (2) for the hcp and bcc phases, respectively. (c) Nucleation rate as a function of local Ni composition. Dashed and solid lines show the interpolated $J(x,T)$ for the hcp and bcc phases, respectively. The inset shows the probability distribution of local Ni concentration, x_{loc}, for bulk liquids with mean compositions of 10 (red) and 20 (orange) at.% Ni. The lines in the inset are Gaussian fit. (d) Macroscopic nucleation rates including compositional fluctuations. Lines represent the baseline nucleation rates without compositional fluctuations, and shaded regions indicate the rates incorporating compositional fluctuations.

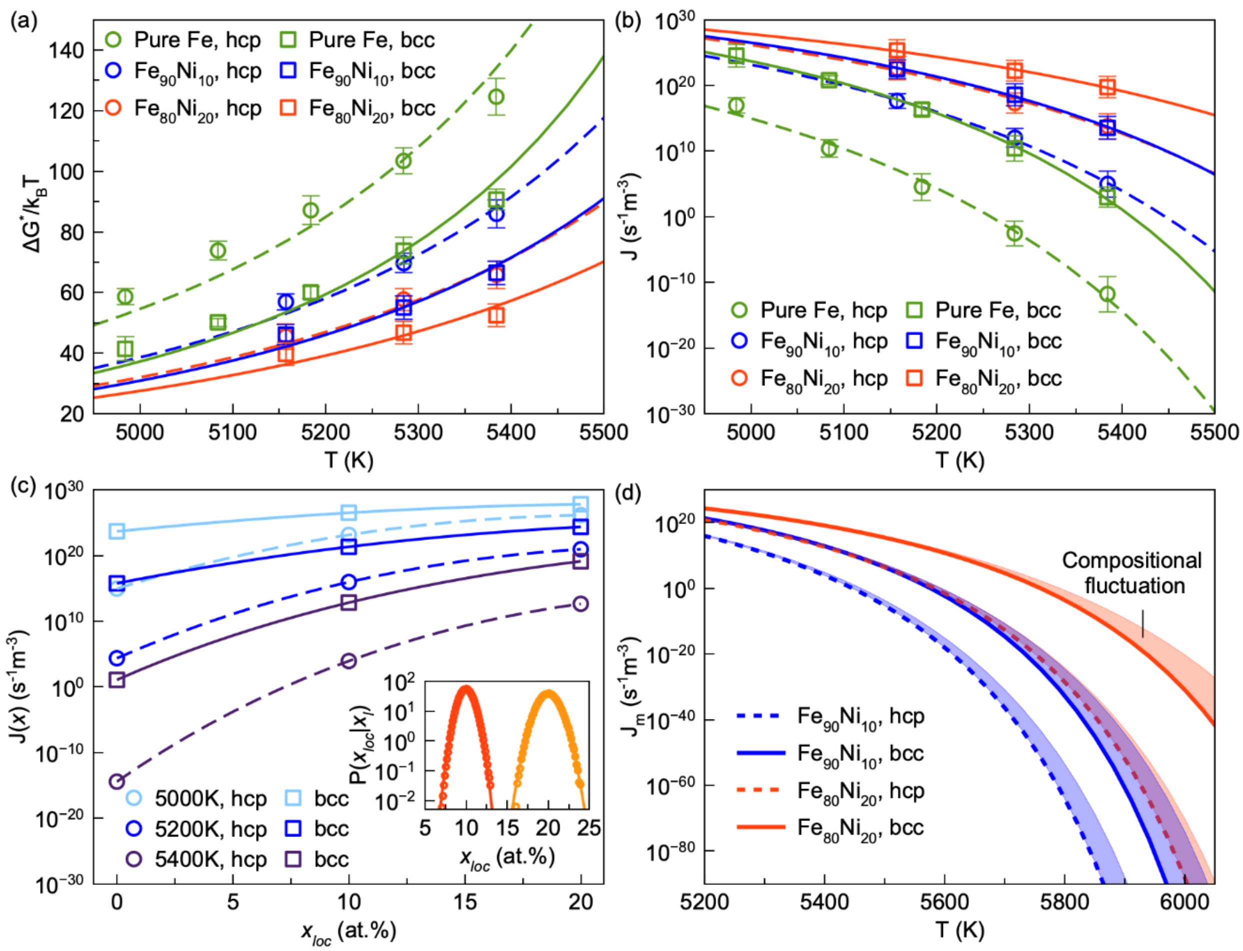
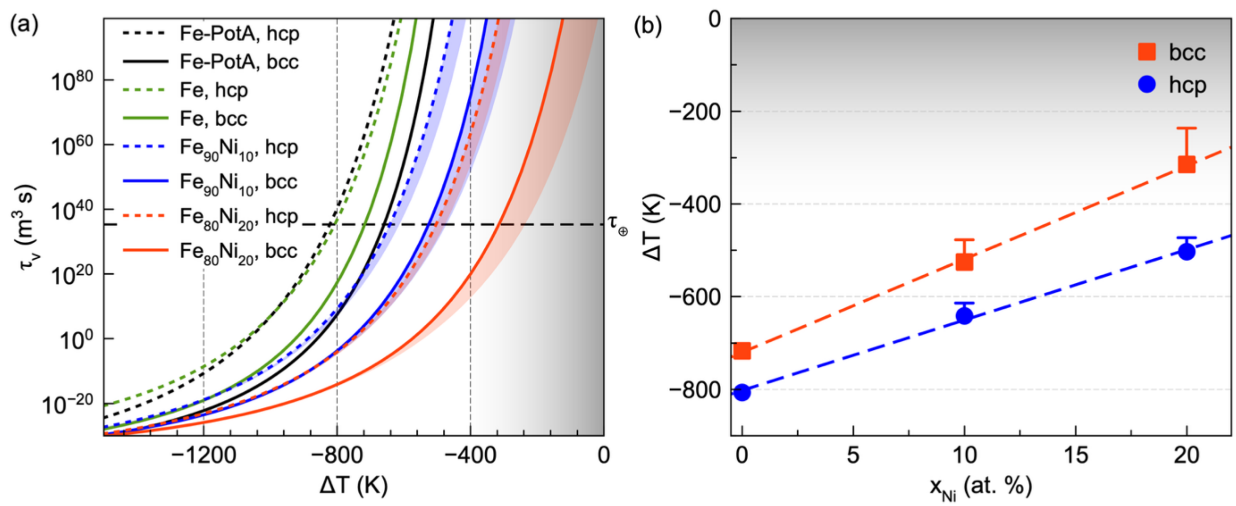


FIG. 4. Nucleation waiting time and supercooling for IC formation. (a) Nucleation waiting time, τ_v, versus supercooling relative to the melting point, ΔT, for Fe and Fe-Ni alloys. Dotted and solid lines represent the hcp and bcc phases, respectively. The horizontal dashed line shows the waiting time for the IC nucleation, $\tau_{\oplus}$. The gray shaded region represents the supercooling range compatible with the current constraints (Huguet et al., 2018; Wilson, Davies, Walker, Pozzo, et al., 2025). The Fe-PotA data are the waiting times from (Gao et al., 2025) computed using Alfe's interatomic potential (Davies et al., 2019). The colored shaded curves indicate the waiting time incorporating compositional fluctuations. (b) Required supercooling to reach $\tau_{\oplus}$ for hcp and bcc nucleation in liquids with different Ni compositions. The bars indicate the effect of compositional fluctuations.



Supporting Information for

Quantifying the effects of nickel on Earth's inner-core nucleation

Jiahui Zhai[1], Liangrui Wei[1], Chen Gao[1], Yang Sun[1]

[1]Department of Physics, Xiamen University, Xiamen 361005, China

Contents of this file

Introduction

This Supporting Information provides additional methodological details and supporting data including

- details of free-energy calculations, persistent-embryo simulations, and interatomic potential
- radial distribution functions for Fe, Ni, and Fe-Ni alloys
- nucleus size and composition change from PEM nucleation simulations
- critical nucleus size and nucleation barrier in Fe80Ni20 liquid
- fitting parameters for the composition-dependent nucleation rate

Text S1. Nucleation Simulations

In classical nucleation theory (CNT) (Debenedetti, 2021) the Gibbs free energy change associated with forming a solid nucleus containing N atoms in a undercooled liquid is written as a competition between a favorable bulk term and an unfavorable interfacial term,

$$\Delta G(N) = -N\,|\Delta\mu| + A(N)\,\gamma, \tag{S1}$$

where $|\Delta\mu|$ is the nucleation driving force. γ is the solid–liquid interfacial free energy per unit area, and $A(N)$ is the nucleus surface area. The competition between the two terms in Eq. (S1) produces a nucleation barrier ΔG_c at a critical size N^*, determined by $\partial\Delta G(N^*)/\,\partial N = 0$. The resulting expression (Sun, Zhang, et al., 2018) is

$$\Delta G_c = \frac{1}{2}\,|\Delta\mu|\,N^*. \tag{S2}$$

Computing the nucleation driving force requires the Gibbs free energy of the alloy system. To this end, we first compute the free energy of the pure Fe and Ni system. Pure Fe liquid iron is chosen as the reference state, setting its Gibbs free energy to zero at given temperature, $G^l_{Fe}(T) = 0$. For the Fe's solid phases, the free energy difference relative to the liquid phase, $\Delta G^{s-l}_{Fe}(T)$ (s denotes bcc or hcp phase), is determined by integrating the Gibbs-Helmholtz equation with respect to temperature starting from melting points, T_m, as

$$\Delta G^{s-l}_{Fe}(T) = -T\int_{T_{m,Fe}}^{T} \frac{\Delta H^{s-l}_{Fe}(T')}{T'^2}\,dT', \tag{S3}$$

where ΔH^{s-l}_{Fe} is the enthalpy difference between the solid phase and the liquid phase. T_m is computed via solid-liquid coexistence simulations (Morris et al., 1994; Sun et al., 2022). For the pure Ni phases, we compute their free energy in a similar way as $\Delta G^{s-l}_{Ni}(T) = -T\int_{T_{m,Ni}}^{T} \frac{\Delta H^{s-l}_{Ni}(T')}{T'^2}\,dT'$.

The Gibbs free energy of the Fe-Ni solutions at a specific Ni concentration x is written as

$$G^{s/l}(x,T,P) = \Delta G^{s/l}_{mix}(x,T,P) + (1-x)G^{s/l}_{\mathrm{Fe}} + xG^{s/l}_{\mathrm{Ni}}, \tag{S4}$$

where $G^{s/l}_{\mathrm{Fe}}$ and $G^{s/l}_{\mathrm{Ni}}$ are the free energy of pure Fe and Ni, respectively. s/l indicates the formula work for both solid and liquid phases. $\Delta G^{s/l}_{mix}$ is the mixing free energy of phase. Here we use the regular solution model to compute the $\Delta G^{s/l}_{mix}$, which has been shown as an accurate model for FeNi alloy under high pressure and high temperature conditions of Earth's core (Wei et al., 2025). In the regular solution model, the mixing free energy is expressed as

$$\Delta G^{s/l}_{mix}(x,T) = \Delta H^{s/l}_{mix}(x) + k_B T[x\ln x + (1-x)\ln(1-x)], \tag{S5}$$

where $\Delta H^{s/l}_{mix}(x)$ is the mixing enthalpy computed via MD simulations.

Because crystal nucleation is a rare event, spontaneous nucleation is often inaccessible to direct MD simulations. The critical nucleus size N^* is computed using the persistent-embryo method (PEM) (Sun, Song, et al., 2018), which was developed to drive the system to overcome the free-energy barrier by introducing a crystalline embryo with an on-the-fly harmonic potential. Once the nucleus reaches its critical size, it has approximately equal probability to grow or shrink, producing a plateau in the nucleus-size trajectory $N(t)$. We compute N^* by averaging $N(t)$ over the plateau regions, and repeat this analysis across independent runs to obtain sufficient statistics.

With N^*, ΔG_c can be computed by Eq. (S2). Then, the nucleation rate J can be obtained by

$$J = \kappa \exp\left(-\frac{\Delta G_c}{k_B T}\right), \tag{S6}$$

where k_B is the Boltzmann constant. κ is a kinetic prefactor. Following steady-state nucleation theory, we use

$$\kappa = \rho_L f^+ \sqrt{\frac{|\Delta\mu|}{6\pi k_B T N^*}}, \tag{S7}$$

where ρ_L is the liquid number density and f^+ is the atomic attachment rate to the critical nucleus. Based on the critical nuclei, we perform iso-configurational ensemble simulations (Widmer-Cooper et al., 2004) to compute the attachment rate f^+, treating it as an effective diffusion constant at the critical nucleus size(Auer & Frenkel, 2004) as

$$f^+ = \frac{\langle[\Delta N^*(t)]^2\rangle}{2t}. \tag{S8}$$

Text S2. Interatomic potential

Due to significant computational cost, it requires an interatomic potential to simulate the nucleation (Gao et al., 2025; Wilson et al., 2023, 2025). Here, we use the embedded atom method (EAM) potentials, which are widely used in the study of alloy systems (Daw & Baskes, 1984; Finnis & Sinclair, 1984). The Fe–Ni interatomic potential used in this work was adopted from our previous study (Wei et al., 2025), with only the Fe–Fe interaction fine-tuned to better match the ab initio results. In the EAM formula, the potential energy is composed of a pairwise term φ and an embedding term F as

$$U = \sum_{i}^{N} \sum_{j=i+1}^{N} \varphi(r_{ij}) + \sum_{i}^{N} F(\rho_i) \tag{S9}$$

where N is the number of atoms in the system. r_{ij} is the distance between atoms i and j, and ρ_i is the charge density on atom i. The function F represents the embedding energy as a function of the local charge density.

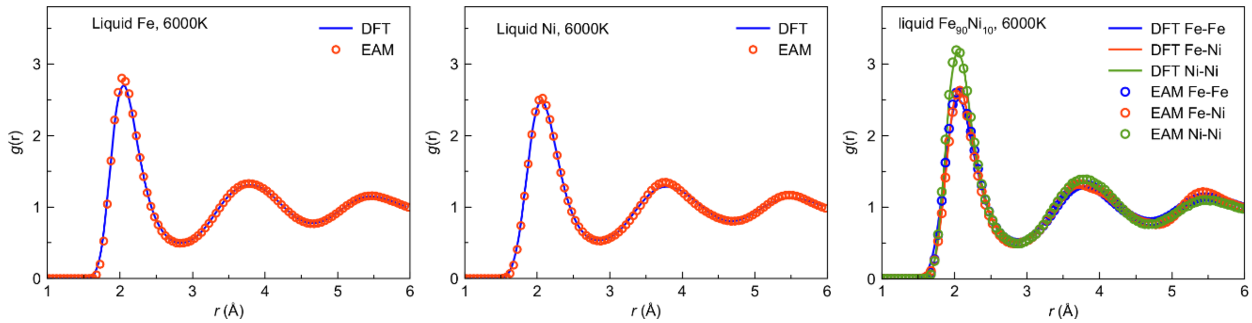


Figure S1 Comparison of radial distribution functions $g(r)$ calculated by the EAM potential and DFT benchmarks at 6000 K for liquid pure Fe (left), liquid pure Ni (middle), and the liquid $\mathrm{Fe_{90}Ni_{10}}$ alloy (right). For the alloy system, the partial RDFs ($g_{\text{Fe-Fe}}$, $g_{\text{Fe-Ni}}$, and $g_{\text{Ni-Ni}}$) are plotted separately. The excellent agreement between the EAM (open circles) and the DFT (solid lines) confirms that the potential effectively reproduces the short-range order and liquid structures at 6000 K, ensuring its reliability for large-scale MD simulations.

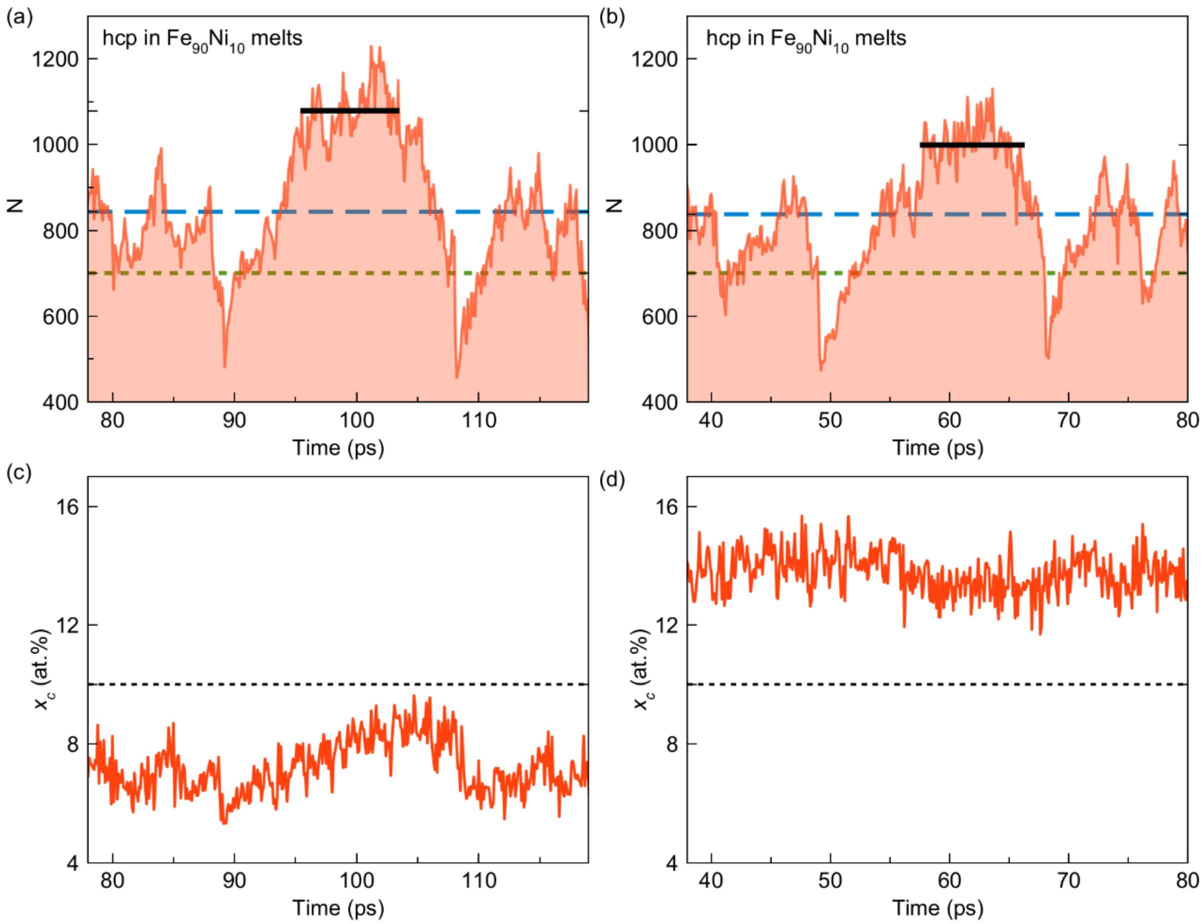


Figure S2 PEM simulation trajectories of the nucleus size N^* and composition x_c for the hcp and bcc phases in $Fe_{90}Ni_{10}$ melts at 5384K. Time evolution of the nucleus size, N^*, in two independent simulation runs (a, b). Corresponding time evolution of the nucleus composition, x_c, for the respective runs (c, d). In (a) and (b), the green dashed line indicates the size of the initial embryo, the blue dashed line shows the N_{sc} for PEM, and the black solid lines highlight the plateau N^* values during the simulation. The black dashed lines in (c) and (d) indicate the nominal melt composition (10 at.%).

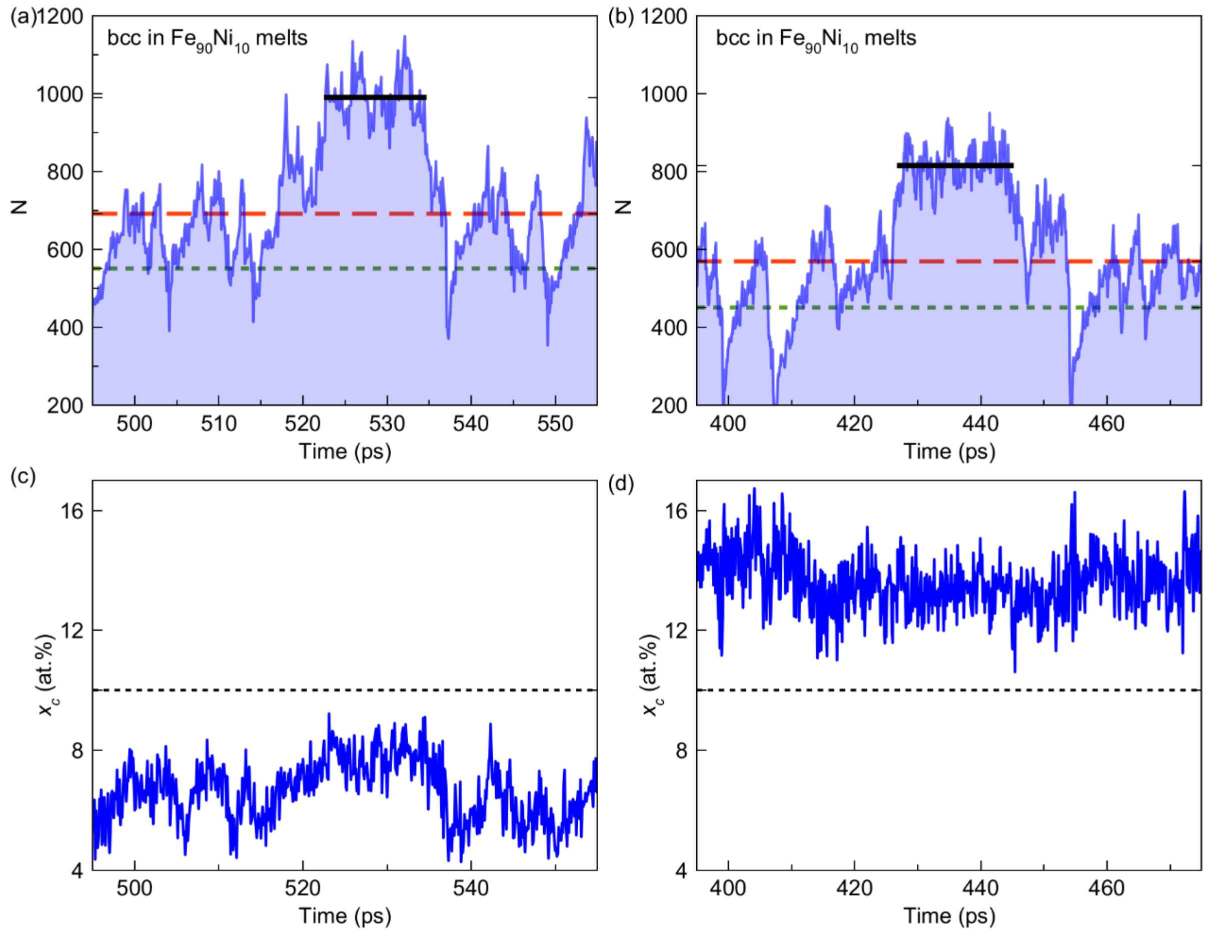


Figure S3 PEM simulation trajectories of the bcc phase in $Fe_{90}Ni_{10}$ melts. The red dashed line shows the N_{sc} and other details are the same as in Fig. S2.

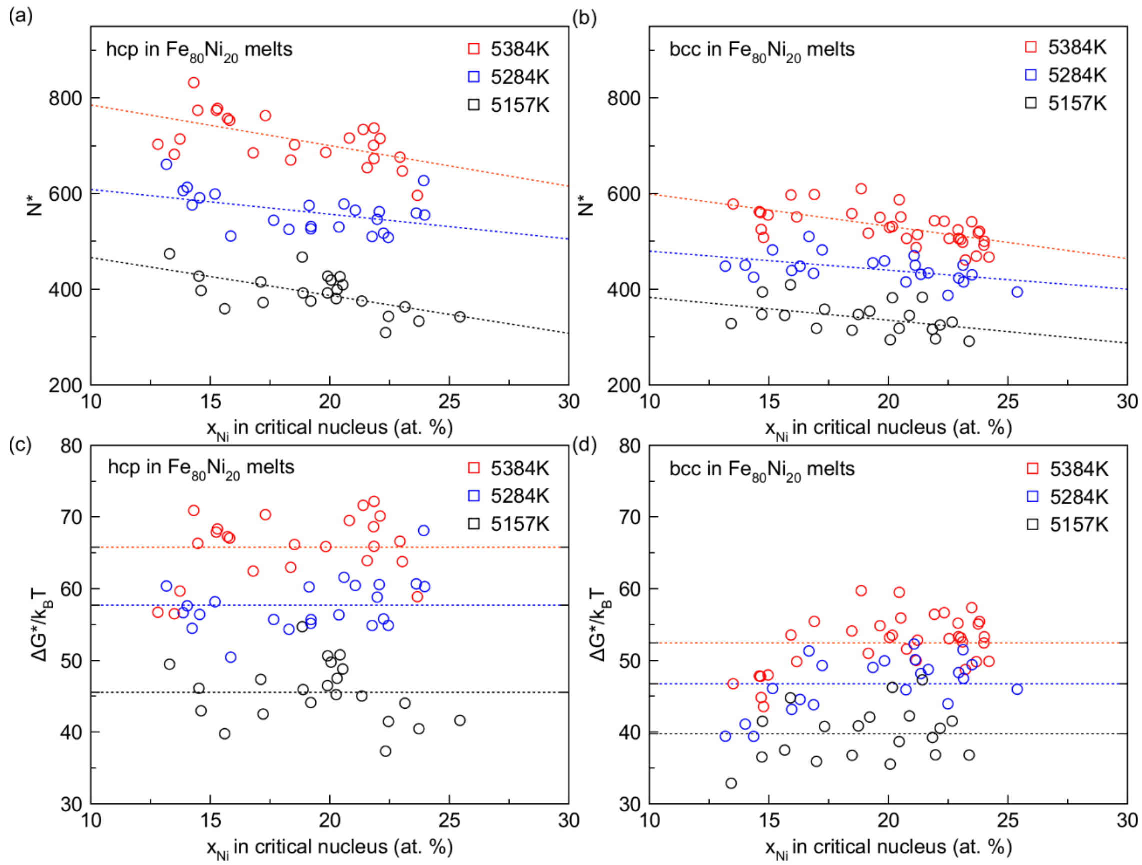


Figure S4 Critical nucleus size and nucleation barrier of $Fe_{80}Ni_{20}$ liquid at 323 GPa. Size, N^*, and composition of the critical nuclei for the (a) hcp and (b) bcc phases. The dotted lines indicate linear fits. Nucleation free-energy barriers for the (c) hcp and (d) bcc phases.

Table S1 Polynomial fitting equations for the composition-dependent parameters of the nucleation rate J in hcp and bcc structures.

	HCP	BCC
B(x)	$0.0007527x^2 - 0.04633x + 3.680$	$-0.0009858x^2 + 0.05066x + 2.195$
$\ln\Gamma(\mathrm{x})$	$-0.03618x^2 + 0.9681x + 84.65$	$-0.01396x^2 + 0.1038x + 89.16$
Tm(x)	$13.00x + 6354$	$27.24x + 6259$
T range	$[4984\ \mathrm{K}, 5384\ \mathrm{K}]$	$[4984\ \mathrm{K}, 5384\ \mathrm{K}]$
x range	$[0.0, 0.2]$	$[0.0, 0.2]$

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